Python in QGIS & ArcGIS

Exercise 4

Goal of this Exercise
In this exercise, you will take your first steps with Python in QGIS. The aim is to explore the various possibilities and write initial scripts that work with geospatial data.

You will work with QGIS Python actions, the console, the editor, as well as Visual Studio Code, and solve various challenges.

When writing python code, please provide meaningful comments.

Materials

Please use the provided date in the folder 'Muenster'. You can download the folder from Learnweb.

Tasks (10 + 5 optional Points)

4.1 – Get knowledge with Wikipedia - Playing with Python actions in QGIS (2 Points)

In the lecture we talked about Python actions in QGIS. With Python actions you can provide smart functionality to work for example with features from a feature layer.

In this task you have to configure a Python action that opens a Wikipedia page in a pop-up window (not a browser!) after clicking on a feature of the shapefile **Muenster_City_Districts**. As you not want to open the general Wikipedia page, your action should open the Wikipedia page for the specific city district.

Please use the classes `QWebView` and `QUrl` to solve this task. You can import them by using

```
from qgis.PyQt.QtCore import QUrl
from qgis.PyQt.QtWebKitWidgets import QWebView
```

In a next step create an instance of the `QWebView` class and use the function `load()` to load your dynamic URL. The URL should be handled by `QUrl` before, e.g. `QUrl("Your URL…")`.

Use the `show()` function of the `QWebView` instance at the end of the script to open the window.

Please hand-in the result as a Python file called **exercise_4_1.py** by uploading it to your repository.

4.2 – Where is the school? – Fun with the QGIS Python editor (4 Points)

In this task, you will use the QGIS Python editor to extract specific information about the dataset **Schools** and print it to the console. You are interested in the name of each school as well as its corresponding X- and Y-coordinates. While the name of the school is already stored in

the column **Name**, the X- and Y-coordinates need to be extracted from the feature's geometry. You can access the geometry using `[your feature variable name].geometry()`. Please find out how to retrieve the X- and Y-coordinates.

To add flexibility to the execution, only the data of the selected feature from the **Schools** dataset should be considered.

The result should be written line by line for each school into a *.csv file named **SchoolReport.csv**. The delimiter should be a **semicolon**. A header line **Name;X;Y** is also necessary. U can use `open('Path to the *.csv', 'w')` to write a csv file.

Please hand-in the result as a Python file called **exercise_4_2.py** by uploading it to your repository.

4.3 MyFirstProject - Create a QGIS Project with content using Visual Studio Code (4 Points)

In this task, you will create a QGIS project from scratch using Visual Studio Code and the sample script **qgis_first_ide_script.py**. This script already contains some code lines to create a new project and to load a shapefile.

Please extend the script in that way that all Shapediles from the **Muenster** folder are stored with their complete path in a list. You can use the `os.listdir()` function for this purpose, which usually returns the result as a list.

Next, you should add the data to the project. Iterate through the list and make sure that only shapefiles are added and that the layer is valid.

The name of the added shapefile in the Table of Contents (TOC) of the project should consist of the original filename of the shapefile without the file extension. You can use the `os.path.basename()` function for this purpose.

Finally, save the project under the name **myFirstProject**.

Please hand-in the result as a Python file called **exercise_4_3.py** by uploading it to your repository.

Please see also the next page!

4.4 Optional challenge - Playing with Geoprocessing tools 😊 (5 Points)

In this task, you should use the **countpointsinpolygon** tool to calculate the number of schools per city district. The input data for this task are the **Schools** and **Muenster_City_Districts** datasets. Use the QGIS Python editor to write and execute the code. You can output the results to the Python console using the `print()` function. A solution could look like this:

```
Innenstadtring: 0.0
Münster-West: 1.0
Münster-Südost: 2.0
Innenstadtring: 4.0
Münster-Hiltrup: 1.0
Mitte-Süd: 2.0
```

Please hand-in the result as a Python file called **exercise_4_4.py** by uploading it to your repository.

General information

The deadline is announced on Learnweb.